

Appendix C

Experimental data for critical heat flux – Low Pressure R123

S/N.	Coil no.	L_h	G	P_{sys}	ρ_l/ρ_g	x_e	Q	CHF
		mm	kg/m ² s	bar			Watt	kW/m ²
1	Coil_1	2146	284	2.11	108.8	1.084	3844	60.0
2	Coil_1	2146	349	2.36	97.2	1.076	4737	74.0
3	Coil_1	2146	422	2.58	88.6	1.073	5765	90.0
4	Coil_1	2146	496	2.82	80.5	1.026	6585	102.8
5	Coil_1	2146	570	3.07	73.7	1.001	7482	116.8
6	Coil_1	1524	436	2.28	100.8	0.588	3538	77.8
7	Coil_1	1524	568	2.59	88.3	0.580	4666	102.6
8	Coil_1	1524	706	2.85	79.6	0.574	5902	129.8
9	Coil_1	1524	867	3.27	68.9	0.515	6905	151.8
10	Coil_1	1524	1012	3.63	61.6	0.498	8078	177.6
11	Coil_1	1006	145	1.74	131.7	1.087	1947	64.9
12	Coil_1	1006	287	1.93	119.0	0.649	2436	81.1
13	Coil_1	1006	425	2.12	108.3	0.523	3060	101.9
14	Coil_1	1006	558	2.27	100.9	0.452	3641	121.3
15	Coil_1	1006	701	2.44	93.8	0.388	4159	138.5
16	Coil_1	1006	710	2.61	87.5	0.303	3641	121.3
17	Coil_1	1006	851	2.59	88.2	0.318	4483	149.3
18	Coil_1	1006	987	2.71	84.0	0.284	4899	163.2
19	Coil_1	504	423	1.85	123.9	0.163	1213	80.7
20	Coil_1	504	560	1.96	117.3	0.136	1483	98.6
21	Coil_1	504	699	2.10	109.6	0.194	2367	157.4
22	Coil_1	504	704	2.06	111.6	0.122	1807	120.1
23	Coil_1	504	851	2.09	109.8	0.098	2001	133.0
24	Coil_1	504	984	2.18	105.3	0.118	2570	170.9
25	Coil_1	504	994	2.13	107.7	0.075	2140	142.2
26	Coil_1	504	1127	2.27	101.3	0.067	2436	162.0
27	Coil_1	504	1276	2.39	95.9	0.055	2682	178.3
28	Coil_1	610	427	1.91	120.4	0.193	1406	77.2
29	Coil_1	610	704	2.14	107.1	0.146	2116	116.2
30	Coil_1	610	983	2.19	104.6	0.137	3030	166.4
31	Coil_1	610	992	2.38	96.4	0.103	2754	151.3
32	Coil_2	1420	432	2.35	97.7	1.003	3513	105.0
33	Coil_2	1420	435	2.36	97.0	1.039	3638	108.7
34	Coil_2	1420	579	2.56	89.3	0.983	4653	139.1
35	Coil_2	1420	707	3.21	70.1	0.886	5359	160.2
36	Coil_2	937	318	1.94	118.8	0.856	2175	98.5
37	Coil_2	937	328	1.97	116.8	0.870	2278	103.2
38	Coil_2	937	370	2.14	107.3	0.897	2700	122.3
39	Coil_2	937	372	2.14	107.5	1.056	3125	141.5
40	Coil_2	937	454	2.23	102.8	0.893	3296	149.3
41	Coil_2	937	510	2.44	93.8	0.786	3357	152.0
42	Coil_2	937	558	2.76	82.3	0.721	3485	157.9
43	Coil_2	937	674	2.69	84.7	0.689	4101	185.8
44	Coil_2	937	775	2.74	83.2	0.619	4334	196.3
45	Coil_2	937	883	2.87	78.9	0.565	4641	210.2
46	Coil_2	937	974	2.98	75.9	0.544	4998	226.4

S/N.	Coil no.	L_h	G	P_{sys}	ρ_l/ρ_g	x_e	Q	CHF
		mm	kg/m ² s	bar			Watt	kW/m ²
47	Coil_2	937	1070	3.27	68.7	0.495	5233	237.0
48	Coil_2	467	206	1.80	127.7	0.718	1195	108.6
49	Coil_2	467	337	1.92	119.8	0.455	1333	121.1
50	Coil_2	467	448	1.74	131.5	0.345	1386	126.0
51	Coil_2	467	457	2.12	108.6	0.331	1435	130.4
52	Coil_2	467	579	2.12	108.2	0.287	1673	152.1
53	Coil_2	467	684	2.25	102.1	0.231	1794	163.0
54	Coil_2	467	825	2.38	96.3	0.196	2003	182.1
55	Coil_2	467	909	2.40	95.3	0.180	2092	190.1
56	Coil_2	467	1130	2.62	87.1	0.138	2350	213.5
57	Coil_2	467	1136	2.15	106.7	0.131	2158	196.1
58	Coil_2	467	1620	2.37	96.7	0.057	2352	213.8
59	Coil_3	1472	421	2.36	97.3	1.169	2103	82.7
60	Coil_3	1472	527	2.62	87.0	1.034	2387	93.8
61	Coil_3	1472	643	2.91	77.9	1.004	2876	113.1
62	Coil_3	1472	694	3.04	74.5	0.972	3042	119.6
63	Coil_3	1472	774	3.28	68.6	0.912	3255	128.0
64	Coil_3	482	435	2.04	112.6	0.615	1172	140.7
65	Coil_3	482	658	2.25	101.8	0.507	1571	188.7
66	Coil_3	482	757	2.37	96.7	0.483	1778	213.5
67	Coil_3	482	845	2.48	92.1	0.458	1950	234.1
68	Coil_3	482	1060	2.81	81.0	0.420	2311	277.5
69	Coil_3	482	1269	3.05	74.0	0.374	2627	315.4
70	Coil_3	482	1466	3.27	68.8	0.328	2861	343.5
71	Coil_3	970	254	1.93	118.8	1.129	1203	71.8
72	Coil_3	970	468	2.46	92.9	0.826	1735	103.5
73	Coil_3	970	647	3.06	73.9	0.772	2355	140.5
74	Coil_3	970	746	3.34	67.2	0.726	2633	157.1
75	Coil_3	970	854	3.68	60.8	0.695	2975	177.5
76	Coil_4	1508	421	2.35	97.3	0.916	4950	110.0
77	Coil_4	1508	498	2.57	88.7	0.813	5333	118.5
78	Coil_4	1508	574	2.72	83.6	0.811	6171	137.1
79	Coil_4	1508	725	3.10	72.8	0.707	7106	157.9
80	Coil_4	989	428	1.85	124.4	0.406	2360	80.0
81	Coil_4	989	564	2.06	111.3	0.357	2899	98.2
82	Coil_4	989	707	2.24	102.6	0.297	3262	110.5
83	Coil_4	989	856	2.41	94.8	0.281	3895	132.0
84	Coil_4	989	1000	2.63	86.7	0.251	4367	148.0
85	Coil_4	989	1002	2.66	85.5	0.255	4451	150.8
86	Coil_4	989	1141	2.71	84.1	0.224	4708	159.5
87	Coil_4	989	1283	2.92	77.7	0.198	5152	174.5
88	Coil_4	506	425	1.84	124.6	0.237	1524	100.9
89	Coil_4	506	578	1.90	121.0	0.188	1775	117.5
90	Coil_4	506	704	2.02	113.7	0.142	1836	121.5
91	Coil_4	506	715	2.00	115.0	0.145	1919	127.1
92	Coil_4	506	853	2.07	110.8	0.130	2182	144.5
93	Coil_4	506	988	2.25	101.8	0.108	2433	161.1
94	Coil_4	506	990	2.24	102.6	0.104	2380	157.6
95	Coil_4	506	1132	2.33	98.3	0.097	2740	181.4

S/N.	Coil no.	L_h	G	P_{sys}	ρ_l/ρ_g	x_e	Q	CHF
		mm	kg/m ² s	bar			Watt	kW/m ²
96	Coil_4	506	1273	2.50	91.5	0.080	3036	201.0
97	Coil_4	506	1274	2.45	93.3	0.075	2928	193.9
98	Coil_5	1529	457	2.40	95.2	0.993	3616	100.4
99	Coil_5	1529	467	2.41	94.8	0.991	3696	102.6
100	Coil_5	1529	568	2.68	84.9	0.929	4296	119.2
101	Coil_5	1529	690	2.99	75.6	0.909	5198	144.3
102	Coil_5	1006	233	1.84	124.6	1.031	1867	78.8
103	Coil_5	1006	241	1.87	122.7	1.026	1928	81.3
104	Coil_5	1006	347	2.10	109.1	0.964	2639	111.3
105	Coil_5	1006	453	2.34	98.0	0.886	3216	135.7
106	Coil_5	1006	573	2.49	91.8	0.753	3583	151.2
107	Coil_5	1006	693	2.65	86.2	0.684	4068	171.6
108	Coil_5	1006	819	2.86	79.2	0.632	4579	193.2
109	Coil_5	1006	941	3.16	71.5	0.643	5423	228.8
110	Coil_5	1006	1078	3.19	70.7	0.585	5897	248.8
111	Coil_5	1006	1195	3.48	64.4	0.528	6317	266.5
112	Coil_5	513	233	1.80	127.4	0.472	919	76.0
113	Coil_5	513	241	1.80	127.8	0.469	946	78.3
114	Coil_5	513	472	2.10	109.4	0.328	1461	120.9
115	Coil_5	513	584	2.21	103.7	0.280	1642	135.9
116	Coil_5	513	696	2.33	98.5	0.250	1851	153.1
117	Coil_5	513	712	2.35	97.6	0.244	1872	154.8
118	Coil_5	513	820	2.44	93.7	0.227	2102	173.9
119	Coil_5	513	929	2.63	86.6	0.214	2393	198.0
120	Coil_5	513	1027	2.79	81.4	0.197	2600	215.1
121	Coil_5	513	1152	2.98	75.9	0.196	3008	248.8
122	Coil_5	513	1385	3.13	72.1	0.149	3272	270.7
123	Coil_5	513	1620	3.30	68.1	0.114	3573	295.6
124	Coil_6	1472	449	2.27	101.0	0.808	1540	61.7
125	Coil_6	1472	469	2.33	98.2	0.848	1685	67.5
126	Coil_6	1472	692	2.64	86.2	0.711	2194	87.8
127	Coil_6	1472	824	3.05	74.1	0.639	2484	99.5
128	Coil_6	1472	946	3.38	66.5	0.572	2688	107.6
129	Coil_6	984	427	2.04	112.8	0.640	1191	71.4
130	Coil_6	984	546	2.23	103.0	0.583	1438	86.1
131	Coil_6	984	675	2.48	92.2	0.561	1775	106.3
132	Coil_6	984	884	2.76	82.6	0.464	2037	122.0
133	Coil_6	984	1092	2.82	80.7	0.426	2426	145.3
134	Coil_6	984	1263	3.28	68.6	0.375	2731	163.6
135	Coil_6	984	1406	3.62	61.7	0.330	2933	175.7
136	Coil_6	476	443	2.19	104.8	0.502	1038	128.6
137	Coil_6	476	661	2.60	87.8	0.363	1277	158.2
138	Coil_6	476	888	2.92	77.7	0.256	1457	180.4
139	Coil_6	476	1182	3.07	73.7	0.200	1765	218.6
140	Coil_6	476	1444	3.27	68.8	0.145	1990	246.4

Appendix D

Experimental data for critical heat flux – High Pressure R123

S/N.	Coil no.	L_h	G	P_{sys}	ρ_l/ρ_g	x_e	Q	CHF
		mm	kg/m ² s	bar			Watt	kW/m ²
1	Coil_1	2146	289	5.36	40.6	1.077	4456	69.6
2	Coil_1	2146	362	5.42	40.0	1.072	5545	86.6
3	Coil_1	2146	437	5.51	39.4	1.031	6506	101.6
4	Coil_1	2146	493	2.76	82.4	1.004	6624	103.4
5	Coil_1	2146	502	5.64	38.4	0.998	7308	114.1
6	Coil_1	2146	581	5.68	38.1	0.990	8408	131.3
7	Coil_1	2146	718	5.73	37.7	0.897	9688	151.3
8	Coil_1	1524	434	5.61	38.6	0.841	5603	123.2
9	Coil_1	1524	579	5.86	36.8	0.866	7650	168.2
10	Coil_1	1524	726	6.13	35.0	0.854	9551	210.0
11	Coil_1	1524	890	6.44	33.3	0.724	10650	234.1
12	Coil_1	1524	975	6.59	32.4	0.678	11273	247.8
13	Coil_1	1524	1069	6.80	31.3	0.654	12186	267.9
14	Coil_1	1006	147	5.06	43.1	0.356	1133	37.7
15	Coil_1	1006	263	5.13	42.5	0.540	2556	85.1
16	Coil_1	1006	436	5.39	40.3	0.401	3711	123.6
17	Coil_1	1006	440	5.24	41.6	0.428	3776	125.8
18	Coil_1	1006	578	5.65	38.3	0.408	4975	165.7
19	Coil_1	1006	719	5.93	36.4	0.475	6699	223.1
20	Coil_1	1006	823	5.87	36.8	0.407	7140	237.8
21	Coil_1	1006	948	5.96	36.2	0.341	7626	254.0
22	Coil_1	1006	1073	6.06	35.5	0.323	8460	281.8
23	Coil_1	610	186	5.22	41.7	0.070	932	51.2
24	Coil_1	610	277	5.24	41.6	0.099	1475	81.0
25	Coil_1	610	439	5.42	40.1	0.392	3701	203.3
26	Coil_1	610	578	5.44	39.9	0.230	3905	214.5
27	Coil_1	610	718	5.47	39.7	0.179	4451	244.5
28	Coil_1	610	854	5.49	39.5	0.107	4635	254.6
29	Coil_1	610	992	5.53	39.2	0.077	5039	276.8
30	Coil_2	1420	483	4.55	48.4	1.020	4235	126.6
31	Coil_2	1420	613	4.65	47.2	1.013	5394	161.2
32	Coil_2	1420	744	4.75	46.2	0.962	6347	189.7
33	Coil_2	1420	904	5.20	41.9	0.729	6398	191.2
34	Coil_2	1420	943	5.37	40.5	0.689	6466	193.2
35	Coil_2	1420	1057	5.46	39.7	0.641	6994	209.0
36	Coil_2	1420	1184	5.56	39.0	0.566	7353	219.8
37	Coil_2	937	343	4.82	45.5	1.008	3042	137.8
38	Coil_2	937	448	4.66	47.1	0.981	3904	176.8
39	Coil_2	937	591	4.86	45.0	0.944	5039	228.2
40	Coil_2	937	675	2.71	84.2	0.434	3037	137.6
41	Coil_2	937	733	5.06	43.1	0.851	5859	265.4
42	Coil_2	937	956	5.45	39.8	0.689	6750	305.7
43	Coil_2	937	1198	5.97	36.1	0.510	7276	329.6
44	Coil_2	937	1631	5.79	37.3	0.278	7365	333.6
45	Coil_2	467	207	4.92	44.5	0.372	998	90.7
46	Coil_2	467	344	5.05	43.2	0.457	1887	171.5

S/N.	Coil no.	L_h	G	P_{sys}	ρ_l/ρ_g	x_e	Q	CHF
		mm	kg/m ² s	bar			Watt	kW/m ²
47	Coil_2	467	479	5.18	42.0	0.360	2378	216.1
48	Coil_2	467	694	5.27	41.3	0.226	2884	262.1
49	Coil_2	467	921	5.22	41.7	0.097	2996	272.3
50	Coil_2	467	1151	5.19	42.0	0.047	3290	299.0
51	Coil_2	467	1356	5.17	42.1	-0.006	3432	311.9
52	Coil_2	467	1568	5.17	42.2	-0.056	3482	316.5
53	Coil_2	467	1569	5.16	42.3	-0.048	3562	323.8
54	Coil_3	1472	378	4.86	45.0	1.097	1939	76.2
55	Coil_3	1472	624	5.61	38.6	1.023	3102	121.9
56	Coil_3	1472	809	5.69	38.0	1.016	3987	156.8
57	Coil_3	1472	1021	6.16	34.9	0.917	4761	187.2
58	Coil_3	1472	1060	3.80	58.6	0.665	3650	143.5
59	Coil_3	1472	1113	6.21	34.6	0.969	5377	211.4
60	Coil_3	1472	1422	6.75	31.6	0.774	6041	237.5
61	Coil_3	970	628	5.54	39.2	0.979	3021	180.2
62	Coil_3	970	820	6.04	35.6	0.897	3769	224.9
63	Coil_3	970	823	6.05	35.5	0.927	3867	230.7
64	Coil_3	970	1136	6.56	32.6	0.718	4607	274.9
65	Coil_3	970	1281	6.56	32.6	0.651	4906	292.7
66	Coil_3	970	1463	6.67	32.0	0.550	5130	306.1
67	Coil_3	482	402	5.02	43.5	0.856	1743	209.2
68	Coil_3	482	636	5.26	41.4	0.660	2344	281.4
69	Coil_3	482	827	5.50	39.4	0.530	2705	324.8
70	Coil_3	482	1091	5.23	41.6	0.365	2890	346.9
71	Coil_3	482	1486	5.48	39.6	0.200	3173	380.9
72	Coil_3	482	1773	5.48	39.6	0.119	3300	396.2
73	Coil_3	482	2062	5.47	39.7	0.070	3497	419.8
74	Coil_5	1529	446	5.03	43.4	1.035	4106	114.0
75	Coil_5	1529	574	5.25	41.5	0.916	4876	135.3
76	Coil_5	1529	704	5.47	39.7	0.943	6130	170.1
77	Coil_5	1529	777	3.18	71.0	0.904	6143	170.5
78	Coil_5	1529	922	5.76	37.6	0.856	7469	207.3
79	Coil_5	1529	1040	5.85	36.9	0.740	7694	213.6
80	Coil_5	1529	1199	5.87	36.7	0.604	7850	217.9
81	Coil_5	1006	461	4.84	45.2	0.959	3998	168.7
82	Coil_5	1006	592	5.05	43.2	0.824	4636	195.6
83	Coil_5	1006	709	5.24	41.5	0.798	5459	230.3
84	Coil_5	1006	941	5.35	40.6	0.576	5977	252.2
85	Coil_5	1006	1182	5.43	40.0	0.486	6891	290.7
86	Coil_5	1006	1378	5.48	39.6	0.379	7067	298.1
87	Coil_5	1006	1610	5.48	39.6	0.305	7428	313.4
88	Coil_5	513	460	4.85	45.2	0.386	2287	189.2
89	Coil_5	513	657	4.99	43.8	0.320	3040	251.5
90	Coil_5	513	659	4.98	43.9	0.299	2955	244.4
91	Coil_5	513	914	5.14	42.4	0.225	3713	307.2
92	Coil_5	513	1138	5.27	41.3	0.162	4203	347.7
93	Coil_5	513	1374	5.16	42.3	0.081	4301	355.8
94	Coil_5	513	1638	5.02	43.5	0.018	4361	360.8
95	Coil_6	1472	409	5.06	43.2	1.096	2023	81.0
96	Coil_6	1472	681	5.38	40.4	0.997	3183	127.5

<i>S/N.</i>	Coil no.	<i>L_h</i>	<i>G</i>	<i>P_{sys}</i>	ρ_l/ρ_g	<i>x_e</i>	<i>Q</i>	<i>CHF</i>
		mm	kg/m²s	bar			Watt	kW/m²
97	Coil_6	1472	928	5.84	37.0	0.944	4204	168.4
98	Coil_6	1472	940	5.80	37.3	0.929	4194	168.0
99	Coil_6	1472	1041	6.00	35.9	0.879	4512	180.7
100	Coil_6	1472	1155	6.17	34.8	0.807	4752	190.3
101	Coil_6	984	391	4.98	43.9	1.025	1841	110.3
102	Coil_6	984	625	5.06	43.1	0.951	2811	168.4
103	Coil_6	984	947	5.63	38.4	0.799	3809	228.2
104	Coil_6	984	1120	6.00	35.9	0.743	4367	261.6
105	Coil_6	984	1175	5.04	43.3	0.817	4734	283.6
106	Coil_6	984	1376	6.03	35.7	0.671	5066	303.4
107	Coil_6	984	1477	6.13	35.0	0.621	5209	312.0
108	Coil_6	984	1719	6.37	33.7	0.565	5800	347.4
109	Coil_6	984	2029	6.72	31.7	0.445	6159	369.0
110	Coil_6	476	360	4.88	44.9	0.769	1400	173.4
111	Coil_6	476	611	5.18	42.1	0.440	1739	215.3
112	Coil_6	476	804	5.38	40.4	0.341	2054	254.4
113	Coil_6	476	1058	5.66	38.3	0.248	2390	295.9
114	Coil_6	476	1062	5.94	36.3	0.230	2390	295.9
115	Coil_6	476	1277	5.43	40.0	0.154	2465	305.3
116	Coil_6	476	1531	5.63	38.4	0.126	2848	352.7
117	Coil_6	476	1765	5.83	37.1	0.095	3143	389.2